Meteo Station — Power Subsystem (generated)

PV−12W −> [U6 INA219] −> CN3791 12V MPPT −> 1S LiPo −> [U7 INA219] −> MT3608 (5.0V) −> DevKit 5V pin.

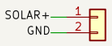
D1 (1N5817): MT3608 5V −> DevKit 5V pin; isolates the boost so USB VBUS wins when flashing (no back−feed).

U6 INA219 0x40: high−side on the PV feed = harvest current + panel voltage.

U7 INA219 0x41: high−side on the battery−>boost feed = load current + battery voltage (Vbus).

Both INA219 on I2C0 (VCC=3V3, 0.1ohm onboard shunt). GPIO2/ADC1_CH1 now a free spare; battery V comes from U7.

J1



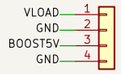
PV−12W 12V 1A

U1



CN3791 12V MPPT

U2



MT3608 −> 5.0V

D1



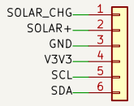
1N5817

U3



ESP32−H2−DevKitM−1

U6



INA219 0x40 (PV)

BT1



1S LiPo 3.7V 10Ah

C1



22uF

U8



BME280 0x76

C2



100uF

C3



10uF

R3



4.75k

R4



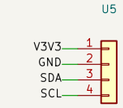
4.75k

U4



BMP388 0x77

U5



MLX90614 0x5A

U7



INA219 0x41 (BATT)

U9



VEML7700 0x10

C5



100nF

C4



10uF

C6



100nF

Capacitor placement (physical):

C1 22uF − bulk at the MT3608 input (VLOAD, right at boost VIN, after the U7 shunt)

C2 100uF − bulk on the 5V rail, at the DevKit 5V pin

C3 10uF − beside C2 at the DevKit 5V pin

C4 10uF − bulk on the 3V3 rail, at the DevKit 3V3 pin

C5 100nF − across BMP388 VCC−GND, hard against the chip's pins

C6 100nF − across MLX90614 VCC−GND, hard against the chip's pins

Rule: a 100nF sits right at each chip; bulk caps go where current steps (boost out, radio, regulator).

U8 BME280 & U9 VEML7700 are breakout modules (onboard decoupling); only the bare MLX90614 needs its external 100nF.

Sheet: /
File: meteo_power.kicad_sch

Title: Meteo Station Power

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.4		Id: 1/1